CS 412: Spring '26
Introduction To Data Mining

Assignment 3

(Due Wednesday, March 25, 23:59)

- The homework is due on **Wednesday, March 25, 2026, at 23:59**. We will be using Gradescope for all homework assignments. You may contact the TAs if you have any technical difficulties while submitting the assignment. Please do NOT email a copy of your solution.
- Please use Campuswire if you have questions about the homework. Make sure to tag your post appropriately. Also, scroll through previous posts to make sure that your query was not answered previously. In case you are sending us an email regarding this Assignment, start the subject with "CS 412 HW3:" and include **all** TAs and the Instructor.
- Make sure that your answers are clear and concise. The homework should be submitted as a single PDF file.

  You are expected to typeset the solutions (using typesetting software such as LaTeX). Handwritten solutions *may not* be graded.
- For a particular question, you will NOT receive full credit if you only provide the final result. Please show all necessary steps and details, including any formulae you use. Clearly describe the variables utilized within the formulae.

  Consider the following two examples.

Example 1 **Q:** Given a dataset $\mathcal{X} = \{3.1, 4.2, -1\}$, compute the mean.
A: For any set of n numbers $\mathcal{X} = \{x_1, \ldots, x_n\}$, the mean can be computed as $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$. For the given dataset $\mathcal{X}$, the mean is $\mu = \frac{3.1+4.2-1}{3} = 2.1$

Example 2 **Q:** A coin claimed to be unbiased has been tossed 100 times, with 54 heads and 46 tails. What is the χ^2 statistic?
A: For a categorical variable taking k possible values, if the expected values are $e_i, i = 1, \ldots, k$ and the observed values are $o_i, i = 1, \ldots, k$, then the χ^2 statistic can be computed as: $S = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{o_i}$. For the problem, since the coin is claimed to be unbiased, the expected values are $50, 50$. Further, the observed values are $54, 46$. Then, the chi-squared statistic is given by $S = \frac{(54-50)^2}{50} + \frac{(46-50)^2}{50} = 0.64$.

- For some problems, writing a short code snippet to perform some computation might be beneficial. However, you are not expected to submit the script. Note that you still have to show all key steps. For instance, a student who states that they used `numpy.mean` (or wrote a short code snippet) for the problem in Example 1 would not get the full points.
- Do NOT change the order of the questions in your submission.
- Start each question on a new page.
- After uploading your submission, Gradescope will prompt you to select the appropriate pages for each subproblem. Make sure that you do this precisely. Unmarked or improperly marked pages *may* lead to a reduction in grade.
- The external datasets referenced within specific problems (if any) can be found on Canvas.

- Late submission policy: there will be a 24-hour grace period without any grade reduction, i.e., Gradescope will accept late submissions until **Thursday, March 26, 2026, at 23:59.** Unfortunately, We will NOT accept late submissions past the grace period (without a reasonable justification).

1. (25 points) The dataset in Table 1 will be used to train a decision tree to predict whether a fruit is **Spoiled** based on its Color, Texture, and Smell.

Color	**Texture**	**Smell**	**Spoiled**
Red	Smooth	1	Yes
Red	Smooth	2	Yes
Yellow	Smooth	1	Yes
Yellow	Rough	1	Yes
Yellow	Rough	2	Yes
Yellow	Smooth	2	No
Yellow	Wrinkled	2	No
Yellow	Rough	3	No
Red	Soft	2	No
Red	Smooth	3	No
Red	Rough	3	No

Table 1: Fruit Characteristics and Spoilage

Answer the following questions:

(a) (11 points) Compute the Information Gain for all of the features. Show all intermediate computations.

(b) (5 points) Draw the full decision tree that would be learned for this dataset (with no pruning and using Information Gain).

(c) (9 points) Now, imagine we add a new feature to the dataset called **Fruit_ID**, which is a unique identifier for each of the 11 fruits in the table (e.g., 'Fruit_1', 'Fruit_2', etc.).

   i. You do not need to do the full calculation, but what would the Information Gain for the `Fruit_ID` feature be? (Hint: Think about the purity of the subsets created by splitting on this feature).

   ii. Based on this, would the standard decision tree algorithm (ID3) choose `Fruit_ID` as the root of the tree?

   iii. Is `Fruit_ID` a useful feature for predicting whether a *new*, unseen fruit is spoiled? Explain why or why not, and describe the fundamental flaw or bias in the Information Gain metric that this example exposes.

Note for 1b

- *Hand drawing is not acceptable. You may use any digital drawing tool to create the sketch as an image and embed it in the Latex file.*

2. (30 points)Linear Models

Part A: Consider the binary classification task illustrated below:

- The horizontal and vertical axes represent two input features.
- The symbols △ and □ denote the two class labels.
- Each plot shows a linear decision boundary learned by a classifier trained using **square loss**.

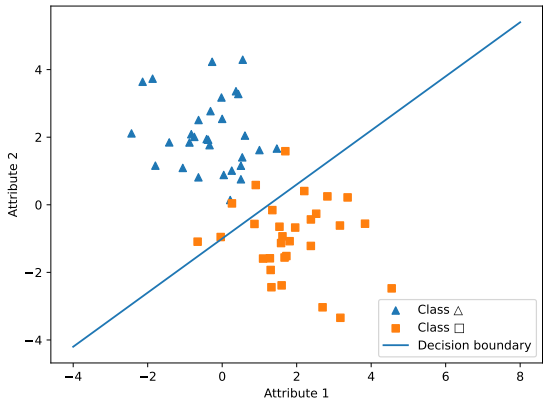

Figure A

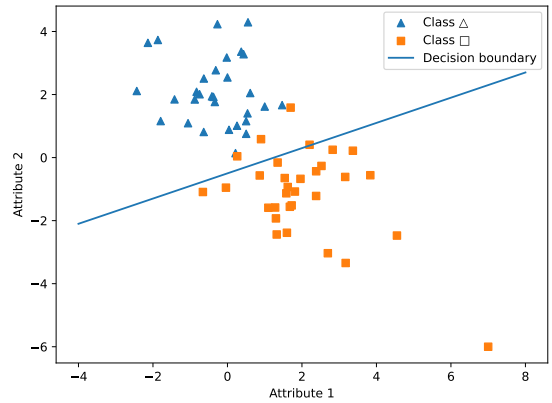

Figure B

(a) (5 pts) Compare the two plots. Specifically, why does the decision boundary in Figure B provide poorer classification performance than the boundary in Figure A?

(b) (5 pts) Explain how the square loss function contributes to the issue observed in the Figure B.

(c) (5 pts) Suppose the extreme point in the right plot were removed. Describe how the learned decision boundary would likely change and explain why.

Part B:

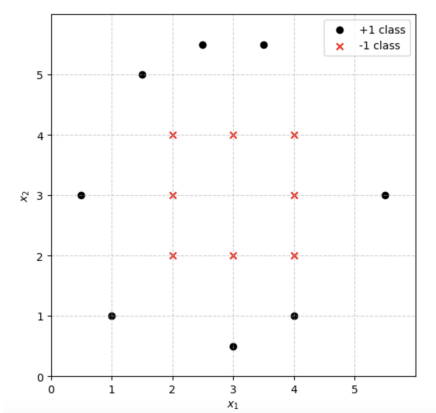

Figure 1: Q??-Dataset 2

(d) (5 points) Describe how you might use an SVM as a classifier to predict whether an item belongs to the + or − class in Figure 1.

(e) (5 points) In SVM, it is often the case that data points cannot be well-separated via a "hard margin". Can you provide a solution to solve this problem?

(f) (5 points) In a soft-margin SVM, the parameter C controls the trade-off between margin width and classification errors.

- What happens when C is very large?
- What happens when C is very small?

3. (25 points) The following dataset summarizes the outcome of a hypothetical *Food Poisoning* predictor given three symptoms: *Nausea*, *Fever*, and *Stomach Cramps*.

Nausea	*Fever*	*Stomach Cramps*	*Has Food Poisoning*
No	No	No	Negative
Yes	No	No	Negative
No	Yes	No	Negative
Yes	No	Yes	Positive
No	Yes	Yes	Positive
Yes	Yes	No	Positive
Yes	Yes	Yes	Positive

Answer the following questions:

(a) (20 points) Fit a Naïve Bayes Classifier to the above dataset. More specifically, compute the likelihood tables for all three features (*Nausea*, *Fever*, *Stomach Cramps*) and fill in the tables below.

		Has Food Poisoning	
		Positive	Negative
Nausea	Yes		
	No		

Table 2: Likelihood Table for *Nausea*

		Has Food Poisoning	
		Positive	Negative
Fever	Yes		
	No		

Table 3: Likelihood Table for *Fever*

		Has Food Poisoning	
		Positive	Negative
Stomach Cramps	Yes		
	No		

Table 4: Likelihood Table for *Stomach Cramps*

(b) (5 points) Using the constructed likelihood tables, classify (perform inference) the following data point as either Positive or Negative: *Nausea* = No, *Fever* = Yes, and *Stomach Cramps* = No. Show all intermediate calculations. Round the final results to four decimals.

4. (20 points) Suppose there are three kinds of candy bags:

- $\frac{1}{4}$ are type h_1: 100% cherry candies,
- $\frac{1}{2}$ are type h_2: 50% cherry candies and 50% lime candies,
- $\frac{1}{4}$ are type h_3: 100% lime candies.

Any candy bag of any type has exactly 100 candies at the beginning.

We have one candy bag, but we do not know which type it is. We want to compute the posterior probabilities of the different types of bags as we draw candies out of the bag we have. The candies are **drawn without replacement**.

(a) (8 points) We draw a candy (Candy_1) from the bag, and it turns out to be "lime." What are the posterior probabilities $p(h_1|\text{Candy}_1 = \text{lime}), p(h_2|\text{Candy}_1 = \text{lime}), p(h_3|\text{Candy}_1 = \text{lime})$ of each type of bag? Clearly explain your answer, e.g., how you are using Bayes rule, and show your calculations. You can use a fraction for final result.

(b) (8 points) We draw another candy (Candy_2) from the bag, and it turns out to be "lime." What are the posterior probabilities $p(h_1|\text{Candy}_1 = \text{lime}, \text{Candy}_2 = \text{lime}), p(h_2|\text{Candy}_1 = \text{lime}, \text{Candy}_2 = \text{lime}), p(h_3|\text{Candy}_1 = \text{lime}, \text{Candy}_2 = \text{lime})$ of each type of bag? Clearly explain your answer, e.g., how you are using Bayes rule, and show your calculations. You can use a fraction for final result.

(c) (4 points) Are your computations in (b) above using conditional independence in any form? If yes, clearly show how it is being used and briefly explain why it is valid. If no, briefly explain why it cannot be used in this problem.